

MT 302: Transport Phenomena

Course Code	MT 302
Course Name	Transport Phenomena
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech. (MSE)
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- **Introduction:** Transport phenomena are vital for understanding the electrical conductivity of metals, semiconductors, and various other fields. Focus on the transport of heat, energy, and mass on a macroscopic scale. Ellipsoidal carrier pockets. Electrons and holes in intrinsic semiconductors. Donor and acceptor doping of semiconductors. characterization of semiconductors. (2 Hours)
- **Thermal Transport:** Thermal conductivity for metals, semiconductors, and insulators. Thermoelectric phenomena in metals. Thermopower for semiconductors. Effect of Thermoelectricity on the thermal conductivity. Thermoelectric measurements. Seebeck Effect (Thermopower). Peltier effect. Thomson effect. The kelvin relations. Phonon drag effect. Electron and Phonon Scattering, Scattering processes in semiconductors. Electron-phonon scattering. ionized impurity scattering. Temperature dependence of the electrical and thermal conductivity. Liouville's equation, Boltzmann equation, carrier scattering, Classical Laws and conservation equations, phonon hydrodynamic equation. (12 Hours)
- **Energy Transport:** Fourier's law and thermal conductivity of materials, gases, solids, liquids and bulk materials. Heat transfer with forced convection in a tube, flat plate, natural convection and the energy equation. Conduction heat transfer: Conduction of heat in solids, the energy equation for conduction, Steady-state one-dimensional systems, Transient systems, finite dimensions, Transient conditions, infinite and semi-infinite solids, Simple multidimensional, Moving sources. Radiation heat transfer: Concept of black body and emissivity, radiation view factors. The energy distribution and the emissive power. Gray bodies and absorptivity, Exchange between infinite parallel plates. Radiation combined with convection. Radiation from gases, transparent solids. Transient conduction with radiation at the surface. Transient heating with thermal stresses. (12 Hours)
- **Mass Transport:** Interphase mass transfer, Definition of fluxes-Fick's first law, Diffusion in solids, Diffusion in ceramic materials, Diffusion in elemental semiconductors, Diffusion in liquids, Diffusion in gases, Diffusion through porous media. Diffusion in solids, Steady-state diffusion experiments. Transient diffusion experiments. Finite system solutions. Microelectronic diffusion processing. Homogenization of alloys. Formation of surface layers. Mass transfer in fluid systems Diffusion through a stagnant gas film. Diffusion in a moving gas stream. Diffusion into a falling liquid film. The mass transfer coefficient. General equation of diffusion with convection. Mass transfer with forced convection over a flat plate. (10Hours)
- **Coupled Fluids with Heat and Mass Transfer:** Coupled fluids, heat and mass transfer, heat and mass transfer coefficients, natural convection (4 Hours)
- **Applications of transport phenomena:** Cooling devices, microfluidics devices, thermoelectric materials, semiconductor/solid-state devices, electrochemical devices, food industry, biomedical. (2 Hours)

Laboratory Modules

1. NA

Text Books

1. Gaskell, David R., **An Introduction to Transport Phenomenon in Materials Engineering**, 2nd Edition, Momentum Press.
2. Bird, R. Byron, W. E. Stewart, and E. N. Lightfoot, **Transport Phenomena**, 4th Edition, Wiley, 2006.
3. Thomson, William J., **Introduction to transport phenomena**, 2000.
4. Robert S., and Harry C. Hershey, **Transport phenomena: a unified approach**, Brodkey Publishing, 2003.

References

1. C. J. Geankoplis, **Transport Processes and Separation Process Principles**, 4th Edition, Prentice-Hall Inc., 2003.
2. C. O. Bennett, J. O. Myers, **Momentum, Heat, and Mass Transfer**, 2nd International Student Edition, McGraw Hill, 1983.